Worked Solutions

Equation, table and figure numbers refer to the *Formula Summary Sheet*; baseline values from the *ASW Example*.

ASW baseline used throughout

$W_{crew} + W_{payload} = 10{,}800\,\text{lb}$; $V = 596.9\,\text{ft/s}$; $C_{cruise}/C_{loiter} = 0.5/0.4\,1/\text{hr}$; $(L/D)_{cruise} = 0.866 \times 16 = 13.86$, $(L/D)_{loiter} = 16$; $W_e/W_0 = 0.93\,W_0^{-0.07}$.
Segment fractions: 0.970, 0.985, 0.8581, 0.9277, 0.8581, 0.9917, 0.995 $\Rightarrow W_7/W_0 = 0.6440$, $W_f/W_0 = 0.3773$, $W_0 = 56{,}706 \approx 56{,}700\,\textbf{lb}$.
Useful constant: Breguet range factor $\dfrac{V\,(L/D)}{C} = \dfrac{596.9 \times 13.86}{0.5/3600} = 5.955 \times 10^7\,\text{ft} = 9{,}801\,\text{nmi}$.

Question 1 — Low-bypass turbofan instead of high-bypass

(a) SFC (Table 3.3). Low-bypass turbofan: $C_{cruise} = 0.8\,1/\text{hr}$, $C_{loiter} = 0.7\,1/\text{hr}$ (was 0.5 / 0.4). V, L/D and payload unchanged.

(b) Segment fractions. The cruise exponent scales with C: $0.153 \times 0.8/0.5 = 0.245$.

No.	Segment	Exponent	Low-bypass	(High-byp.)
3, 5	Cruise, Eq. (6)	$\dfrac{(1500 \times 6076)(0.8/3600)}{596.9 \times 13.86} = 0.245$	$e^{-0.245} = 0.7828$	(0.8581)
4	Loiter 3 hr	$3 \times 0.7/16 = 0.131$	$e^{-0.131} = 0.8770$	(0.9277)
6	Loiter 20 min	$\frac{1}{3} \times 0.7/16 = 0.0146$	$e^{-0.0146} = 0.9855$	(0.9917)

$$\frac{W_7}{W_0} = 0.970 \times 0.985 \times 0.7828 \times 0.8770 \times 0.7828 \times 0.9855 \times 0.995 = \mathbf{0.5035} \quad (\text{was } 0.6440)$$

(c) Fuel fraction and sizing. $\dfrac{W_f}{W_0} = 1.06\,(1 - 0.5035) = \mathbf{0.5263}$ (was 0.3773)

$$W_0 = \frac{10{,}800}{1 - 0.5263 - 0.93\,W_0^{-0.07}}$$

W_0 guess	W_e/W_0	W_0 calc.
150,000	0.4038	154,478
160,000	0.4020	150,558
153,000	0.4032	153,252
153,200	0.4032	153,171
153,180	0.4032	**153,179**

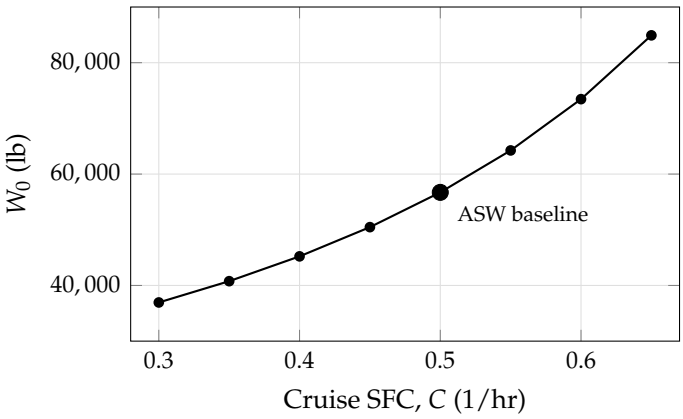
$W_0 \approx 153{,}200\,\text{lb}$ (vs 56,700 lb)
$\approx 2.7\times$ heavier, $+170\%$
$W_e \approx 61{,}800\,\text{lb}$, $W_f \approx 80{,}600\,\text{lb}$

(d) Comment. A 60% (cruise) / 75% (loiter) rise in SFC almost *triples* W_0. The fuel fraction rises exponentially with C (Eqs. 6–7), and the denominator of Eq. (1), $1 - W_f/W_0 - W_e/W_0$, collapses from 0.191 to 0.071: the payload is now only 7% of take-off weight. More fuel needs more structure, which needs more fuel — the sizing "snowball". Engine SFC is therefore one of the strongest drivers of aircraft size; low-bypass engines suit fighters/supersonic aircraft, not long-endurance patrol aircraft.

Question 1 — Extension: continuous SFC trade study

Scaling *both* cruise and loiter SFC by a factor k and resizing each case gives the trade curve below (useful if the class finishes early).

k	C_{cr}	C_{lt}	W_3/W_2	W_4/W_3	W_7/W_0	W_f/W_0	W_0 (lb)	ΔW_0
0.6	0.30	0.24	0.9123	0.9560	0.7526	0.2623	36,944	−34.9%
0.7	0.35	0.28	0.8984	0.9489	0.7238	0.2927	40,767	−28.1%
0.8	0.40	0.32	0.8848	0.9418	0.6962	0.3220	45,223	−20.3%
0.9	0.45	0.36	0.8713	0.9347	0.6696	0.3502	50,470	−11.0%
1.0	0.50	0.40	0.8581	0.9277	0.6440	0.3773	56,715	—
1.1	0.55	0.44	0.8451	0.9208	0.6194	0.4034	64,245	+13.3%
1.2	0.60	0.48	0.8322	0.9139	0.5958	0.4285	73,456	+29.5%
1.3	0.65	0.52	0.8196	0.9071	0.5730	0.4526	84,911	+49.7%

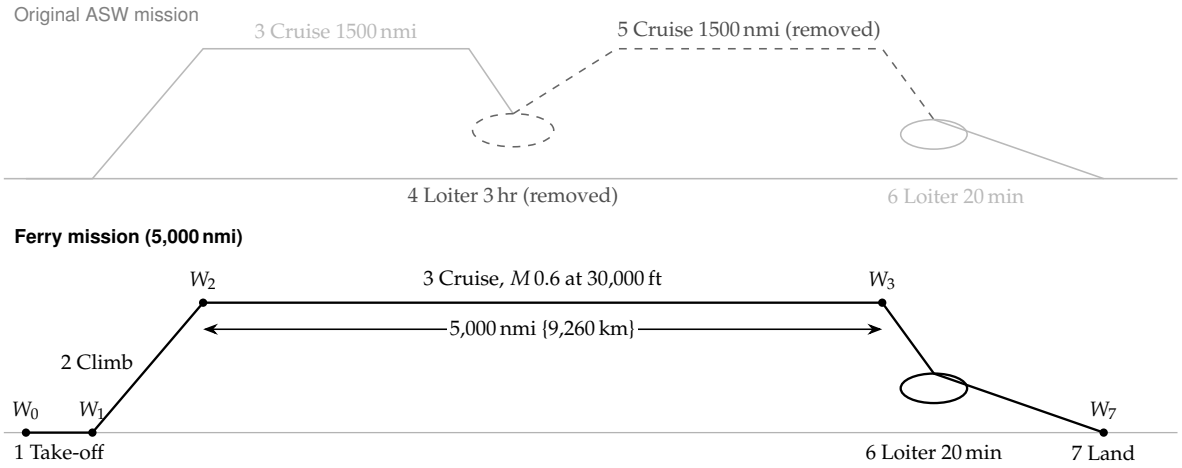


Findings

- W_0 grows *non-linearly* with SFC: +10% SFC → +13% W_0, but −10% SFC → −11%.
- The "snowball": more fuel → bigger structure → more fuel, so the curve steepens.
- Around the baseline, $\partial W_0/\partial C \approx 1.4 \times 10^5$ lb per 1/hr (i.e. ≈ 1,400 lb per 0.01).
- SFC is one of the strongest levers on aircraft size.

Question 2 — Resize for a 5,000 nmi ferry mission

Mission: 1 take-off → 2 climb → 3 cruise 5,000 nmi → 6 loiter 20 min → 7 land (segments 4 and 5 removed). Crew and avionics (10,800 lb) are assumed to remain on board.



Segment 3 (Eq. (6)): the exponent scales directly with range,

$$\frac{RC}{V(L/D)} = \frac{(5000 \times 6076)(0.5/3600)}{596.9 \times 13.86} = 0.153 \times \frac{5000}{1500} = 0.510 \quad \Rightarrow \quad \frac{W_3}{W_2} = e^{-0.510} = 0.6004$$

Mission fraction (Eq. (4)) and **fuel fraction** (Eq. (5)):

$$\frac{W_x}{W_0} = 0.970 \times 0.985 \times 0.6004 \times 0.9917 \times 0.995 = 0.5660, \qquad \frac{W_f}{W_0} = 1.06\,(1 - 0.5660) = 0.4600$$

Sizing (Eq. (1), Eq. (2)): $W_0 = \dfrac{10{,}800}{1 - 0.4600 - 0.93\,W_0^{-0.07}}$

W_0 guess (lb)	W_e/W_0	W_0 calculated (lb)
80,000	0.4220	91,489
90,000	0.4185	88,880
89,000	0.4188	89,120
89,100	0.4188	89,096
89,097	0.4188	**89,097**

$$W_0 \approx 89{,}100\ \text{lb}\ \{40{,}400\ \text{kg}\}, \quad W_e \approx 37{,}300\ \text{lb}, \quad W_f \approx 41{,}000\ \text{lb}$$

Compared with the ASW mission (56,700 lb), the ferry mission needs a ≈ 57% heavier aircraft: one 5,000 nmi leg costs far more fuel than two 1,500 nmi legs plus a 3 hr loiter, because the fuel fraction rises exponentially with range. In practice a ferry mission is flown by the *existing* aircraft with external/ferry tanks rather than by sizing a new one.

Question 3 — Effect on range of a 20% SFC reduction

(a) Quick estimate. From Eq. (8), $R \propto 1/C$ for the same weight fraction, so each cruise leg becomes $1500/0.8 = 1875$ nmi: total 3,750 nmi (+25%).

(b) Full answer — same aircraft, same fuel. $W_0 = 56{,}706$ lb, W_e and W_f are unchanged, so $W_7/W_0 = 0.6440$ is fixed. New SFC: $C = 0.4$ (cruise), 0.32 (loiter).

Loiter fractions now use less fuel:

$$\frac{W_4}{W_3} = e^{-3\times 0.32/16} = e^{-0.060} = 0.9418, \qquad \frac{W_6}{W_5} = e^{-(1/3)\times 0.32/16} = e^{-0.00667} = 0.9934$$

Fuel left for the two cruise legs:

$$\frac{W_3}{W_2}\cdot\frac{W_5}{W_4} = \frac{0.6440}{0.970\times 0.985\times 0.9418\times 0.9934\times 0.995} = \frac{0.6440}{0.8894} = 0.7241 \;\Rightarrow\; \text{each leg } \sqrt{0.7241} = 0.8510$$

Range of each leg (Eq. (8)):

$$R = \frac{V}{C}\frac{L}{D}\ln\frac{W_{i-1}}{W_i} = \frac{596.9\times 13.86}{0.4/3600}\ln\frac{1}{0.8510} = 12{,}251 \text{ nmi} \times 0.1614 = 1{,}977 \text{ nmi}$$

Total cruise range $= 2 \times 1{,}977 \approx 3{,}950$ nmi, i.e. $\approx +32\%$ (vs 3,000 nmi)

The gain exceeds the simple $1/0.8 = +25\%$ because the 3 hr and 20 min loiters also burn less fuel, releasing extra fuel for cruise. (Alternatively, keeping the mission fixed and resizing gives $W_0 \approx 45{,}200$ lb, a 20% lighter aircraft — see Q1, $k = 0.8$.)

Question 4 — Range if the aircraft weighs 50,000 lb

Weights. With the same payload and empty-weight trend (Eq. (2)):

$$\frac{W_e}{W_0} = 0.93\,(50{,}000)^{-0.07} = 0.4361 \;\Rightarrow\; W_e = 21{,}803 \text{ lb}$$

$$W_f = W_0 - W_e - (W_{crew} + W_{payload}) = 50{,}000 - 21{,}803 - 10{,}800 = 17{,}397 \text{ lb} \;\Rightarrow\; \frac{W_f}{W_0} = 0.3479$$

Mission fraction from Eq. (5): $\dfrac{W_7}{W_0} = 1 - \dfrac{0.3479}{1.06} = 0.6718$

Cruise fraction available (loiter and other segments unchanged):

$$\frac{W_3}{W_2}\cdot\frac{W_5}{W_4} = \frac{0.6718}{0.970 \times 0.985 \times 0.9277 \times 0.9917 \times 0.995} = \frac{0.6718}{0.8747} = 0.7680 \;\Rightarrow\; \text{each leg } \sqrt{0.7680} = 0.8764$$

Range of each leg (Eq. (8)):

$$R = 9{,}801 \text{ nmi} \times \ln\frac{1}{0.8764} = 9{,}801 \times 0.1320 = 1{,}293 \text{ nmi}$$

Total cruise range $= 2 \times 1{,}293 \approx 2{,}590$ nmi {4,790 km}, i.e. $\approx 14\%$ short of 3,000 nmi

A 12% lighter aircraft carries the same 10,800 lb payload but only 17,400 lb of fuel (vs 21,400 lb), so the radius of action falls by 14% while the 3 hr loiter is retained.

Past Exam Question 2(a) — GA trainer: cruise and loiter weight fractions

Step 1 – Wetted-area ratio (Fig. 3.6). A single-engine tractor with a straight, high-aspect-ratio wing and conventional tail is closest to the *Cessna Skylane RG*: $S_{wet}/S_{ref} \approx 4.0$.

Step 2 – Wetted aspect ratio (Eq. (10)). $A_{wet} = \dfrac{A}{S_{wet}/S_{ref}} = \dfrac{8}{4.0} = 2.0$

Step 3 – $(L/D)_{max}$ (Fig. 3.5). "Retractable" places it on the *retractable prop aircraft* curve; at $A_{wet} = 2.0$: $(L/D)_{max} \approx 15.5$ (15–16 acceptable).

Step 4 – L/D per segment (Table 3.5, prop). $(L/D)_{cruise} = (L/D)_{max} = 15.5$, $\quad (L/D)_{loiter} = 0.866 \times 15.5 = 13.4$

Step 5 – SFC (Table 3.4, piston-prop fixed pitch; Eq. (9)). $V = 70\,\text{m/s} = 229.7\,\text{ft/s}$

$$C_{cruise} = \frac{0.4 \times 229.7}{550 \times 0.8} = 0.209\ 1/\text{hr} = 5.80 \times 10^{-5}\ 1/\text{s}, \qquad C_{loiter} = \frac{0.5 \times 229.7}{550 \times 0.7} = 0.298\ 1/\text{hr}$$

Step 6 – Cruise fraction (Eq. (6)), $R = 1000\,\text{km} = 10^6\,\text{m}$:

$$\frac{RC}{V(L/D)} = \frac{10^6 \times 5.80 \times 10^{-5}}{70 \times 15.5} = 0.0535 \quad \Rightarrow \quad \left.\frac{W_i}{W_{i-1}}\right|_{cruise} = e^{-0.0535} = \mathbf{0.948}$$

Step 7 – Loiter fraction (Eq. (7)), $E = 30\,\text{min} = 0.5\,\text{hr}$ (loiter speed taken as $70\,\text{m/s}$):

$$\frac{EC}{L/D} = \frac{0.5 \times 0.298}{13.4} = 0.0111 \quad \Rightarrow \quad \left.\frac{W_i}{W_{i-1}}\right|_{loiter} = e^{-0.0111} = \mathbf{0.989}$$

Cruise: $W_i/W_{i-1} \approx 0.948$ Loiter: $W_i/W_{i-1} \approx 0.989$

Points to stress: (i) for a *propeller* aircraft, cruise uses $(L/D)_{max}$ and loiter $0.866(L/D)_{max}$ — the reverse of a jet; (ii) C must be converted with Eq. (9) using V in ft/s, and then to 1/s if R and V are in SI; (iii) no loiter speed is given, so state the assumption — the propeller C depends on V. Sensitivity: $(L/D)_{max} = 15 \to 16$ gives cruise 0.946–0.950 and loiter 0.9886–0.9893.